Bennett Fix

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EDUCATION

University of Pittsburgh, Swanson School of Engineering — Pittsburgh, PA
B.S. Computer Engineering, Concentration in Autonomous Systems, Minor in Philosophy — *Expected Aug 2028*

- **Coursework:** Embedded Processors and Interfacing, Data Structures and Algorithms, Signals and Systems
- **Registered, Spring 2027:** Cyber-Physical Systems, Embedded Systems Design, Computer Organization and Architecture, Applied Discrete Mathematics, Junior Design

PROJECTS

Virtual Attraction Control System (VACS) | *C++, CMake, Catch2* — Completed Nov 2026 | GitHub

- Developed a multi-vehicle attraction control simulator implementing block occupancy management, dispatch authorization, and station safety interlocks
- Added show control that triggers lighting, audio, and effects cues as vehicles reach each scene
- Implemented fault-handling logic for sensor failures, communication faults, and emergency-stop conditions, preventing vehicle movement when required safety conditions are not met
- Wrote automated Catch2 unit tests to verify dispatch behavior, fault responses, and control logic

RC Car | *STM32, 12-24 V brushed DC motor* — *In progress*

- Building an RC car from the ground up around an STM32 microcontroller; completed first drive motor test

Elevator Control System | *ARM Cortex-M4 assembly* — GitHub

- Wrote an elevator controller in assembly that handles keypad floor selection, tracks car position, drives a stepper motor between floors, and shows the current floor on a 7-segment display

STM32 ASCII Animation | *C, STM32 Nucleo*

- Programmed an STM32 Nucleo board to play an ASCII art animation over UART to a serial terminal

Pathfinding and Maze Solving | *C++*

- Implemented A* search and a BFS maze solver, along with custom list and queue data structures

EXPERIENCE

Walt Disney World, Attractions Cast Member (Disney College Program) — Aug 2026 - Present
Slinky Dog Dash, Disney's Hollywood Studios — *Orlando, FL*

- Operate Slinky Dog Dash following established ride safety procedures, dispatch protocols, and operational requirements
- Perform assigned operational safety checks and coordinate with fellow Cast Members to support safe, efficient operation

Lavner Education, STEM Instructor — Summer 2026

- Taught Python and introductory neural network concepts to beginner students around age 11
- Built Tkinter and pygame projects (classifiers, games) written for students to read and modify
- Reworked curriculum that was too advanced for the group, including a scaffolded MNIST classifier in Google Colab

Athena Coding, Instructor — Summer 2025

- Learned Lua and taught it to campers

SKILLS

- **Hardware:** STM32, FPGA, ARM Cortex-M4, stepper motors, brushed DC motors, UART, keypads, 7-segment displays
- **Languages:** C, C++, ARM assembly, VHDL, Python, Lua
- **Development and testing:** Git, GitHub, CMake, VS Code, STM32CubeIDE, Catch2 unit testing, Quartus, ModelSim
- **Libraries:** Tkinter, pygame, Google Colab

LEADERSHIP AND ACTIVITIES

Tech Behind the Dreams, Disney

- Selected to participate in Disney's Tech Behind the Dreams experience, gaining exposure to the technical disciplines involved in themed entertainment

Eagle Scout, Scouts BSA Troop 127

- Served as Senior Patrol Leader, leading the troop's youth leadership team